

Job-to-Job Flows in the Great Recession[†]

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Labor market conditions have devolved rapidly since the recession began in December 2007. More than two years following its official end in June 2009, unemployment remains persistently high and the duration of unemployment historically long. Underlying this surge in joblessness is a rich degree of heterogeneity in labor market outcomes as workers and firms adjust to the crisis, which existing public use statistical datasets only partially address. In this paper, we construct and examine new measures of labor market adjustment, the flows of workers between jobs, including both direct job-to-job flows and separations that involve nonemployment, to generate a fuller portrait of labor market adjustment during the Great Recession.

Numerous studies, including Jacobson, Lalonde and Sullivan (1993), have documented persistent wage declines following job separations, while a separate literature that includes Topel and Ward (1992) has documented that job change is an important component of early career wage growth. An emerging literature on job-to-job flows, including Fallick and Fleishman (2004); Golan, Lane, and McEntarfer (2007); and Bjelland et al. (2011), ties these literatures together. As job flows and wage changes are rooted in the same process, an understanding of job-to-job flows is key for understanding employment and earnings dynamics, as well as the winners and losers from the expansion and contraction of different industries.

We utilize a multistate pilot database of job separations to calculate the frequency of different types of job-to-job flows, along with

associated earnings changes for the years 1998–2010. We document that direct job-to-job flows fell sharply in the Great Recession, while greater shares of workers separated to long nonemployment spells. We find that earnings changes associated with job change are procyclical, with marked penalties for nonemployment. Earnings changes for all types of job change are at a series low in the Great Recession, with greater penalties associated with nonemployment in this period. We also take a closer look at labor market adjustment in a principal industry affected by the housing bust, residential building construction. We find a drop in flows across employers and an increase in the rate of industry change and earnings loss.

I. National Job-to-Job Flow Statistics

Measures of transitions across jobs and industries are one of the major opportunities for development of new employment statistics. The Census Bureau plans to develop a new set of national job-to-job flow statistics derived from the Longitudinal Employer-Household Dynamics (LEHD) infrastructure files. These linked employer-employee data, described in Abowd et al. (2006), provide a unique opportunity to trace the flows of workers across employers, industries, and geographies.

The results described here are tabulated on a recently constructed multistate pilot database. The literature on job-to-job flows informed the construction of our series in several important ways. Bjelland et al. (2011) report that most direct job-to-job flows are between primary (maximum earnings) jobs, and so we restrict our analysis here to flows between primary jobs. Fallick, Haltiwanger, and McEntarfer (2011) show that nonemployment is associated with lower between-job wage changes. The importance of nonemployment in earnings outcomes influenced our inclusion of job-to-nonemployment-to-job flows in addition to direct job-to-job flows.

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We use data on separators from 1998–2010 to generate measures of job-to-job and job-to-nonemployment-to-job transitions between primary jobs, as well as separations to longer term nonemployment.¹ Note that in the administrative data we cannot distinguish between those who are unemployed and those not in the labor force. Furthermore, nonemployment durations are only approximately observed in quarterly earnings data. For example, a worker with one full quarter of nonemployment between jobs has a nonemployment spell of three to eight months.

II. Job-to-Job Flows and the Business Cycle

Business cycle changes in labor turnover are important because declines in the rate at which workers are reallocated across jobs impacts the efficiency of the labor market. As job change is an important component of earnings growth over worker careers, declines in labor turnover also have implications for lifetime earnings. Young workers entering labor markets in recessions might experience lower wage growth in part because of lower gains from job change. Figure 1 shows seasonally adjusted primary job separations along with job-to-job and job-to-nonemployment-to-job flows for our LEHD sample from 1998–2010. Separations are modestly procyclical, with a much sharper decline in the Great Recession than in the 2001 recession. Job-to-job flows occurring within the same quarter and those where the new job began in the subsequent quarter demonstrate very similar cyclical patterns and so are combined in Figure 1.

Figure 1 demonstrates several interesting facts about job-to-job flows and their relation to worker flows generally. First, both job-to-job flows involving little or no nonemployment are procyclical, while job-to-nonemployment-to-job spells that involve at least one full quarter of

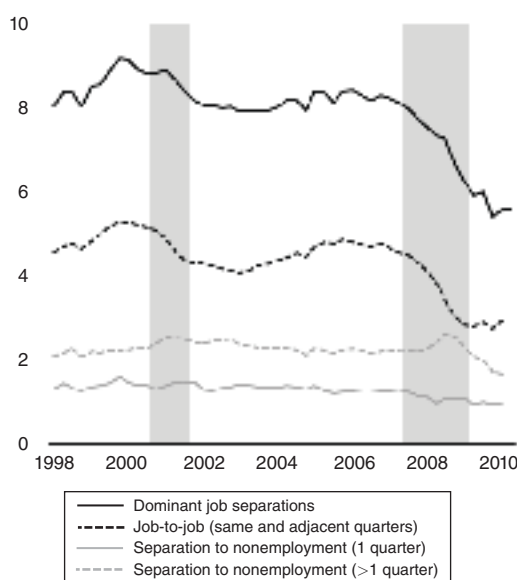


FIGURE 1. QUARTERLY EMPLOYMENT FLOWS (Millions)

Notes: Shaded areas indicate quarters of NBER recessions. Data are seasonally adjusted.

nonemployment demonstrate either no cyclical pattern or, in the case of those involving longer nonemployment spells, a slight countercyclical pattern. Thus the procyclicality of dominant job separations appears to be driven entirely by those first two types of job flows. These two types of flows begin to fall in early 2007, preceding the official start of the Great Recession by a quarter or two and drop to a series (at least 12-year) low by early 2009. The spike in separations to longer nonemployment spells in late 2008 is driven almost entirely by a spike in separations to nonemployment spells lasting one year or more (not shown), consistent with the persistently high unemployment rate and long unemployment durations during this recession.

Figure 2 shows seasonally adjusted median earnings changes from job change for the time period.² Similar to Fallick, Haltiwanger, and McEntarfer (2011), we find that earnings changes associated with job change decrease

¹ In Hyatt and McEntarfer (2012) we provide systematic definitions for our job-to-job and job-to-nonemployment-to-job measures. We differ from Bjelland et al. primarily in that we construct flows between primary jobs, and we include flows to jobs that include a nonemployment spell. We use nine states as the frame for our analysis: California, Florida, Georgia, Illinois, Kansas, Michigan, Nevada, North Carolina, and North Dakota. Specifically, the frame for our analysis is all workers who held at least one job in these nine states during this time period. We then construct national job histories for these workers so that flows for more than 40 states are included.

² As we can compare earnings only for jobs where we observe earnings for the entire quarter, all of our earnings results are for those job transitions where both the origin and destination job survive the quarter, thus excluding transitions between very short jobs.

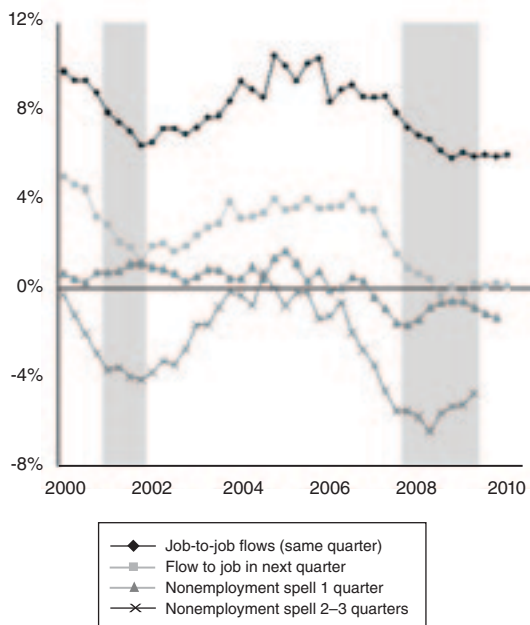


FIGURE 2. PROPORTIONAL CHANGE IN REAL EARNINGS

Notes: Shaded areas indicate quarters of NBER recessions. Data are seasonally adjusted.

with both the presence and duration of a nonemployment spell between jobs. For example, in the second quarter of 2006, workers with direct job-to-job flows experienced a 9 percent earnings gain, those with flows to a new job starting in the following quarter experience a 3.8 percent earnings gain, while those with one or two-to-three quarters nonemployment experience a 0 percent and -1.2 percent earnings change, respectively. We also find procyclicality in earnings changes associated job flows that include a nonemployment spell. What is perhaps most interesting in Figure 2 is the procyclical comovement of earnings changes associated with most types of job flows. There is also some evidence here of an increased “penalty” for nonemployment in the Great Recession; while earnings gains for direct job-to-job flows are similar to the last recession’s, earnings losses are greater for those with two to three quarters of nonemployment.

III. Labor Market Adjustment and the Housing Bust

One of the most interesting applications of a job-to-job flows series is the ability to examine

how labor associated with a particular industry adjusts to a demand shock. In this section we provide an examination of the recent downsizing of the residential building construction industry (North American Industry Classification System 2361). The decadelong boom in housing generated enormous demand for labor in construction. With the collapse of the housing market beginning in 2006, employment in the residential construction industry declined sharply.

Table 1 shows employment and earnings outcomes for residential construction separators who experienced less than a full quarter of nonemployment. The third column of Table 1 shows a sharp decline in both job-to-job flows that originate from residential construction and the share of those flows that are to another employer in the construction sector in the Great Recession. Relative to the 2004–2006 period, in 2007–2009, job-to-job flows decline to about 70 percent of their previous level, and conditional on taking place, nearly two-thirds of flows are to destinations outside construction. Earnings gains associated with job change also decline markedly in the Great Recession. Movements within the residential construction industry are associated with wage gains in the 2001–2006 period, but the median earnings change in the most recent recession is -0.6 percent. Median earnings change associated with movement to another industry is also lower. In 2007–2009, the median earnings change is -3.4 percent, but the two prior periods have small positive earnings changes. This mostly reflects a decline in median earnings across sectors, but it is also the result of more moves to low-wage sectors: among all NAICS sectors, the largest increase between 2004–2006 and 2007–2009 is a more than 2 percentage point increase in the share of flows into the Accommodation and Food Services sector, which are associated with large (13 percent–21 percent) downward movements in earnings.

IV. Conclusion

This article serves two purposes. First, it demonstrates how measures of flows across jobs constructed from linked employer-employee data can shed light on employment and wage dynamics and provide new information about the economy. Second, it provides some new evidence on labor turnover and earnings dynamics in the Great Recession.

TABLE 1—JOB-TO-JOB FLOWS ORIGINATING IN RESIDENTIAL BUILDING CONSTRUCTION

	Frequency of destination			Wage change (median)		
	2001–03	2004–06	2007–09	2001–03	2004–06	2007–09
Construction	43.2	41.4	36.5	5.1	6.9	1.1
Residential building construction	15.6	14.9	11.0	3.9	5.2	–0.6
Other construction industry groups	27.6	26.6	25.4	6.2	8.4	2.7
Sectors other than construction	56.8	58.6	63.5	1.7	2.7	–3.4
Admin., suppt., and waste mgmt.	13.3	14.5	13.3	–0.8	0.5	–3.9
Retail trade	7.6	7.3	7.7	0.0	0.0	–10.6
Accommodation and food services	7.7	7.9	10.0	–17.8	–13.5	–20.2
Manufacturing	5.2	5.4	5.0	10.5	10.3	4.8
Prof., sci., and tech. services	3.2	3.4	4.0	6.5	6.9	2.5
Health care and social assistance	2.4	2.2	3.1	–0.1	4.1	–3.4
Other services (except publ. admin.)	2.3	2.4	3.0	–2.4	–2.6	–6.6
Wholesale trade	2.3	2.5	2.8	8.3	8.7	2.2
Agric., forestry, fishing, and hunting	1.7	2.0	2.7	–3.6	–1.4	–8.1
Any other sector	11.1	10.9	11.9	4.3	4.8	–1.2
Job-to-job flows (thousands)	455.3	650.9	463.5	118.5	191.4	142.4

Notes: Calculated from the set of all job-to-job flows which are within quarter or in adjacent quarters, in which the origin industry is in Residential Building Construction (NAICS Industry Group 2361). Associated median wage changes are available for the subset of job-to-job flows in which both the separation is from and accession is to full-quarter employment; see text for details. Wage changes are calculated for full-quarter earnings of separation job *S* and accession job *A* according to $(A - S)/((A + S)/2)$.

We find evidence that the fall in dominant job separations is driven primarily by a fall in direct job-to-job flows and those with very short intervening spells of nonemployment. We find procyclicality in earnings changes associated with job change. Together these indicate that both job mobility and earnings gains associated with job mobility are on a marked decline in this recession. Taking a close look at one particular industry, residential construction, we find evidence that even separators who did not experience nonemployment have experienced earnings losses, due in part to much higher rates of industry change in this recession.

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